Problem 1.3

The current interest rate quoted by a bank on its savings accounts is 9% per year. You open an account with a deposit of $1,000. Assuming there are no transactions on the account such as depositing or withdrawing during one full year, what will be the amount value in the account at the end of the year?

Solution

Let the initial deposit be
$$PV = 1000$$
and let the annual interest rate be
$$r = 9\% = 0.09.$$

After one year, the accumulated amount is
$$A = PV(1 + r).$$
Substituting the given values,
$$A = 1000(1 + 0.09).$$
Therefore,
$$A = 1000(1.09) = 1090.$$
Hence, the amount in the account at the end of the year is
$$\boxed{\$1{,}090}.$$